4/08/25

Task 2 A

generating design of Other Traditional Database model

Aim

To Study the evolution of traditional database models and their design apporoaches

Procedure

1. Identify models: Hierarchical (tree), Network (graph), Relational.
2. Study Structure parent-child, record Links, table with key
3. Compare features: data Organization, flexibility, ease of use
4. Example: Hierarchical -> Company - Dept -> Employee etc...

1. CREATE TABLE

Definition used to Create a new table in database

Query: Sql

```
CREATE TABLE Employee (
 EmpID INT,
 EmpName VARCHAR (100),
 Department VARCHAR (50),
 Salary INT
);
Sql
CREATE TABLE Department (
 DeptID INT,
 DeptName VARCHAR(50),
 Location VARCHAR(50)
);
```

Output:

Tables Employee and Department Created Successfully

2. Describe or DESC

Display the Structure of a table (Column names and data types)

Query:

Sql

```
DESC Employee;
```

After updating stu table

Select from Student

S.NO	Stu.ID	stu.Name	Stu.depa	Stu.gender
1	28587	k.pavan	CSE	male
2	29847	Sai	EEE	male

S.NO	Stu.ph.no	stu.depar.ID
1	868807789	1225
2	904346789	1425

2 Output

Field	Type
EmpID	INT
EmpName	VARCHAR(100)
Department	VARCHAR(50)
Salary	INT

3 DROP TABLE

Deletes the entire table structure and all its data

Query

Sql

DROP TABLE Employee;

Output

Table Employee dropped successfully

4 ALTER TABLE

used to add, delete, or modify columns in an existing table

Query:

Sql

ALTER TABLE Employee ADD joiningDate DATE;

Output

Column joiningDate added to employee.

UPDATE Student
SET Email: vtu29833@gmail.com
where Roll No: 1;

SELECT * From Students;

SNo	R.No	Name	age	Course	Email
1	"	Pavan	19	B.Tech	vtu29833@gmail.com
2	3	joy	21	B.Tech	joy@gmail.com

Select Name from student;

S.No	Name
1	Sivavan
2	joy

Select * from Students
where Name = 'pavan'.

S.No	Roll	Name	Age
1	1	pavan	19

VEL TECH	
Score	21
PERFORMANCE (vtu29a@g)	5
RESULT AND ANALYSIS	5
VIVA VOCE	5
RECORD (5)	—
TOTAL (20)	15
SIGN WITH DATE	

21/3/25

Result

Thus, the evolution of traditional database models and their design approaches has been implemented successfully

Task - 2.2 11/01/25

DDL AND DML COMMANDS WITH CONSTRAINTS

Aim:

To execute and understand DDL and DML Commands in SQL along with INSERT INTO, SELECT, UPDATE, DELETE, SELECT with WHERE Clause

procedure:

- Create database and table using DDL Commands with Suitable Constraints.
- Alter the table if required using ALTER TABLE
- Insert records into the table using DML Commands while following Constraints
- update and delete records using UPDATE AND DELETE
- Retrieve data using SELECT to Verify results

1 DDL Commands for Hospital Management System

1.1 CREATE TABLE patient (

```
patientID INT PRIMARY KEY,
PatientName VARCHAR(50) NOT NULL,
gender CHAR(1) CHECK (gender IN ('M','F')),
Age INT CHECK (Age >0),
Contact Number VARCHAR(15) UNIQUE,
Address VARCHAR (100)
);
CREATE TABLE Doctor(
DoctorID INT PRIMARY KEY,
DoctorNAme VARCHAR(50) NOT NULL,
Specialization VARCHAR (50),
PhoneNumber VARCHAR (15) UNIQUE
);
CREATE TABLE Appointment (
AppointmentID INT PRIMARY KEY,
patientID INT NOT NULL;
DoctorID INT NOT NULL,
AppointmentDate DATE DEFAULT CURRENT-DATE,
Diagnosis VARCHAR (200),
```

SELECT FROM STUDENT → Before

S NO	Student	STU. DEPADTM	STU. gender	ST. pho. no
			male	98484632
1	Ravi ~~Chandu~~	102	female	98483847
2	Anitha ~~Nishika~~	101		

Select * from Department. Before performing Alter Command

SNO	Stu	Stu. Name	dept	gender	email
				male	Null
1	101	Ravi	102		vdtech@gmail.com
2	102	Anvitha	101	Female	velfech@gmail.com
3	103	Pavan	101	male	

Select * from student after performing Command

S.NO	Stu. ID	Name	St_depart	stud.gen	PNO	Email.
1	1	Ravi	102	Male	98487638	NULL
2	2	Anitha	101	Female	98487665	NULL

FOREIGN KEY (patient ID) REFERENCES patient (patient ID).
FOREIGN KEY (Doctor ID) REFERENCES Doctor (Doctor ID)

INSERT DATA

INSERT INTO Patients (patientID, patientName, gender, Age, Email, Contact
VALUES (1, 'john Doe', 'M', 35, '9876543210', 'New Delhi', 'john@gmail.com')
INSERT INTO Doctor (DoctorID, 'DoctorName, Specialization, phonenumber)
VALUES (101, 'Dr. Meera sharma', 'Cardiologist', '9123456789');
INSERT INTO Appointment (AppointmentID, patientID, DoctorID,
AppointmentDate, Diagnosis)
VALUES (1001, 1, 101, '2025 08-15', 'Mild chestpain');
Result
Records inserted Successfully.

UPDATE Data
UPDATE patient
SET Age = 36, Adress = 'Mumbai'
WHERE PatientID = 1;
Result: john Doe's age updated to 36 and address changed to Mumbai

DELETE DATA
DELETE FROM Appointment
WHERE AppointmentID = 1001;
Result - Appointment with ID 1001 deleted.

SELECT DATA
SELECT p.patientName, d. DoctorNAME, a. AppointmentData, a. Diagnosis
FROM Appointment,

VEL TECH	
EX NO	2.2
PERFORMANCE	5
RESULT	7
VIVA	3
RECORD	—
TOTAL	15

Result:
thus, DDL and DML Commands in ~~SQL along~~ with INSERT, UPDATE, DELETE and SELECT is implemented Successfully.
4/8/25